Week 2 Report

Content Generation & AI Productivity

AI for Content Creation & Automation

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Programme:	CAPACITI — AI Skills Development
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Project Link:	https://tlotliso-ai-generator.netlify.app/

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1. Theme & Overview

Week 2 of the CAPACITI programme focused on **AI for Content Creation & Automation**. The week was designed to build practical fluency in working with generative AI models, moving beyond simply using AI tools to actually engineering prompts, structuring outputs, and integrating AI into a productive workflow. The week culminated in the design and deployment of an individual content-generation project that demonstrates these skills end to end.

The reporting period covered **27 March to 1 April 2026**, and emphasised individual ownership: each learner was expected to identify a real-world content problem, prototype a solution using an AI model, refine it through prompt iteration, and publish the result as part of their portfolio.

2. Core Competencies

The following core competencies were developed and assessed during the week:

- **Prompt optimization** — crafting, testing and refining prompts to consistently achieve high-quality, on-brief outputs from generative AI models.
- **AI content generation** — producing usable text, code and structured content with large language models such as Google Gemini.
- **Individual project ownership** — scoping, building, deploying and documenting a complete project independently.
- **AI productivity workflows** — embedding AI into day-to-day workflows so that drafting, reviewing and iterating happen faster and with higher quality.

3. Technical Concepts Covered

- Prompt iteration and refinement — running controlled variations of a prompt and comparing outputs to identify what wording, context and structure work best.
- AI-generated text, images and code — understanding the strengths, limits and appropriate use cases for each output type.
- Content structuring using AI — using the model to produce outlines, sections, summaries and structured data such as JSON or markdown.
- Integrating AI APIs into a web application — calling the Google Generative Language API (Gemini) from a deployed front-end.

- Responsible use of AI — disclosing AI involvement, validating output, and avoiding the publication of unchecked content.

3. Coursera Courses Completed

To support the practical work, the following Coursera courses were completed during the week and contributed directly to the project deliverable:

- **Prompt Engineering for ChatGPT** — Vanderbilt University.
- **Generative AI with Large Language Models** — DeepLearning.AI & AWS.
- **Introduction to Generative AI** — Google Cloud.

5. Project Deliverable — AI Content Generator

The Week 2 deliverable is a fully working **AI content generator** that takes a user prompt and returns structured, ready-to-use content. The application is built as a single-page web app, calls the Google Generative Language API (Gemini) for inference, and is deployed publicly via Netlify.

Live application:

https://tlotliso-ai-generator.netlify.app/

Key features:

- Prompt-driven content generation for blogs, marketing copy, emails and short-form posts.
- A built-in **prompt library** with reusable, optimised prompt templates.
- Clean, responsive UI for entering prompts and reviewing generated output.
- Direct integration with the Gemini API endpoint (generativelanguage.googleapis.com/v1beta/models/gemini).
- Continuous deployment via Netlify so updates ship automatically from source.

6. Prompt Engineering Case Study

A central part of the week was a documented prompt-engineering case study built around the content generator. The case study followed a simple loop: write an initial prompt, review the model output, identify weaknesses (vague tone, missing structure, hallucinated facts), then refine the prompt with clearer role, context, constraints and output format.

Across multiple iterations, the prompts evolved from short, ambiguous instructions into structured prompts that specified the audience, tone, length, format and forbidden patterns. This produced markedly more consistent, on-brief output and is preserved in the project's prompt library for reuse.

7. Portfolio Development

- Uploaded the content generator project as a featured portfolio piece.
- Added the prompt-engineering case study with before/after prompt examples.
- Expanded the overall portfolio structure to clearly separate projects, case studies and learning artefacts.

8. Measurable Outcomes

Outcome	Status
Functional content generation tool	Achieved — deployed on Netlify
Demonstrated prompt optimization	Achieved — case study + prompt library
Individual project published	Achieved — public live URL
Coursera courses completed	Achieved — 3 courses

9. Project Screenshots & Evidence

The screenshots below provide visual evidence of the deliverable, including the live user interface, the prompt library, the underlying source code, the Netlify deployment, and the Google Generative Language API integration.

https://generativelanguage.googleapis.com/v1/models?key=AIzaSyDqUJ5gu-kLocRVnk95CpCmKEBljqJ7HS4

https://learn-eu-ce... Gmail YouTube Maps Acer New Tab Amazon Booking.com DStv - Watch Live s...

Pretty-print


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Figure 1 *— Live AI Content Generator interface (tlotliso-ai-generator.netlify.app).*

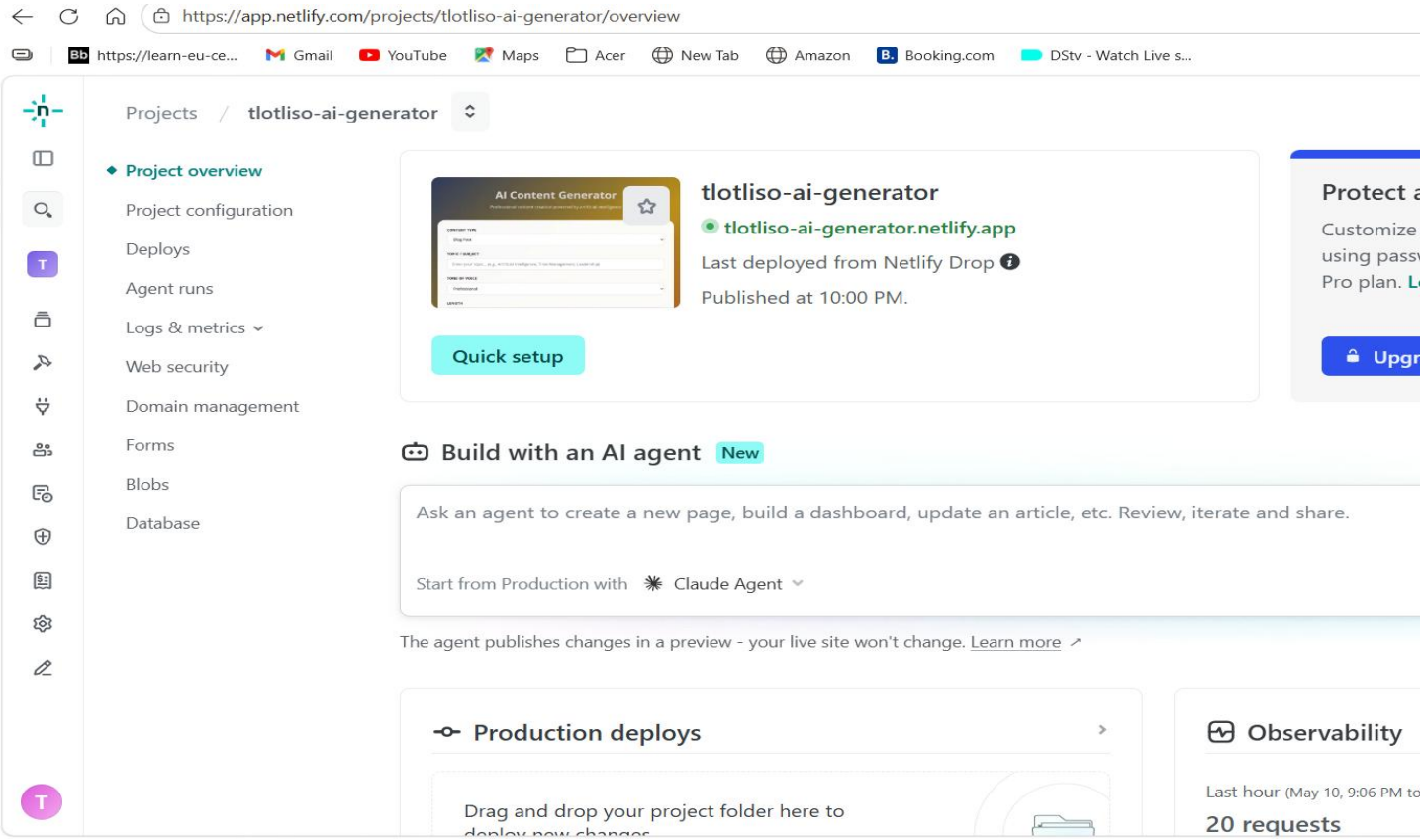


Figure 2 — Built-in prompt library with reusable, optimised prompt templates.

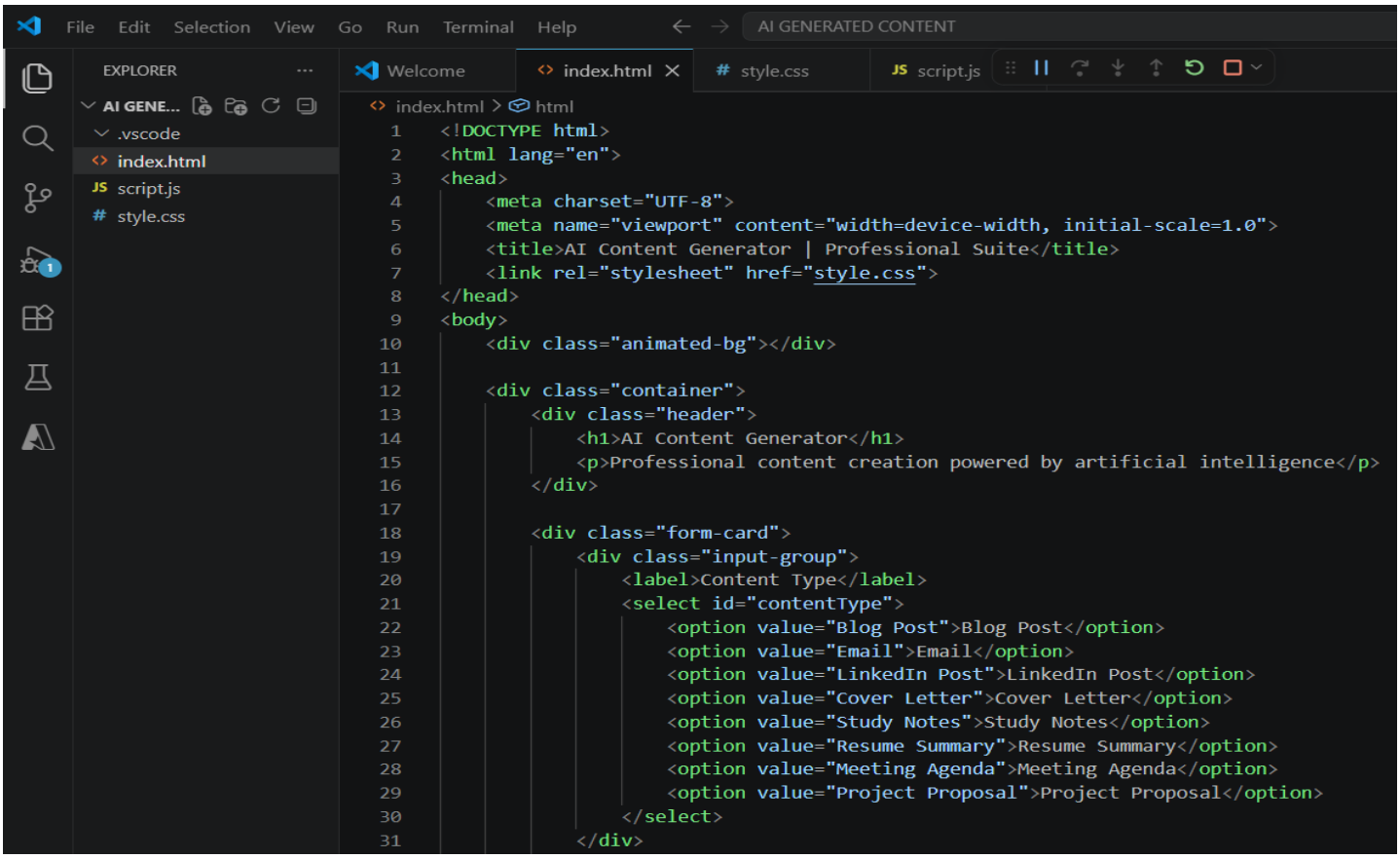


Figure 3 — Project source code structure.

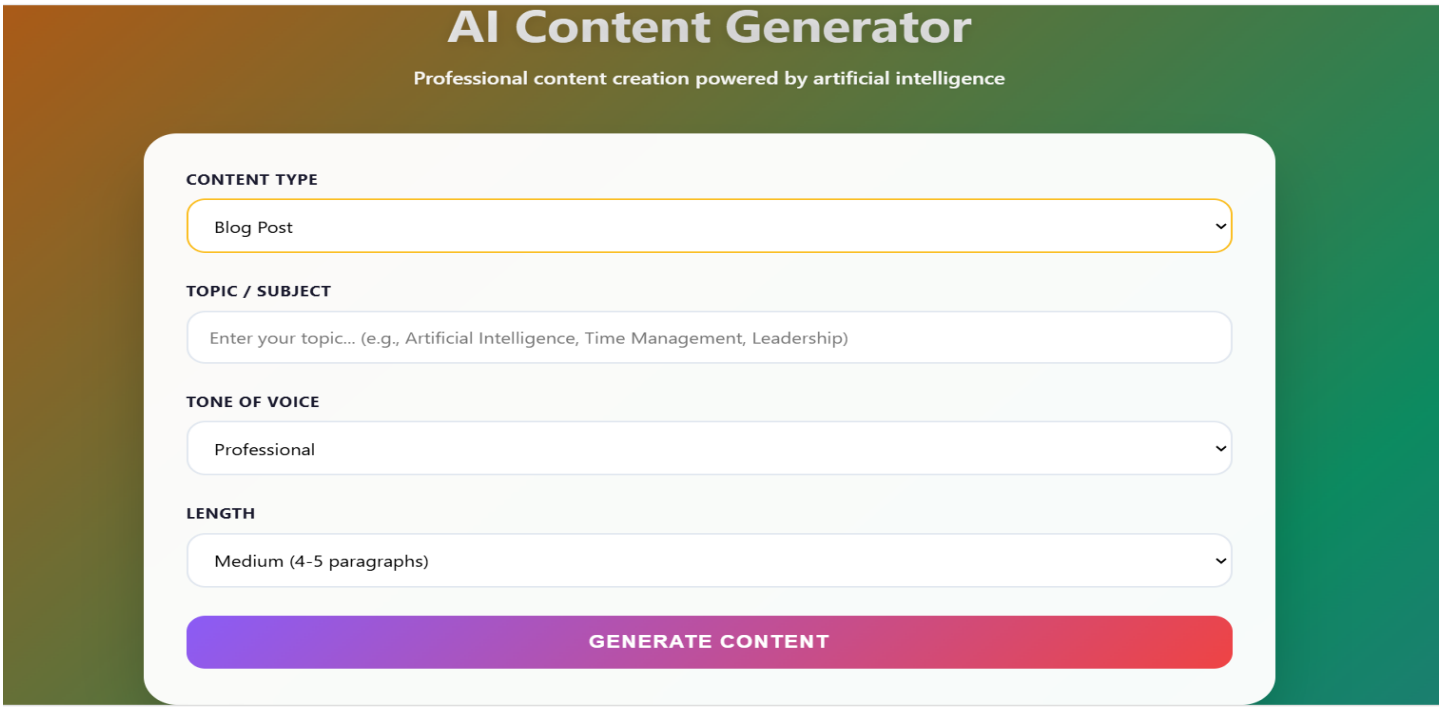


Figure 4 — Netlify deployment dashboard confirming successful publish.

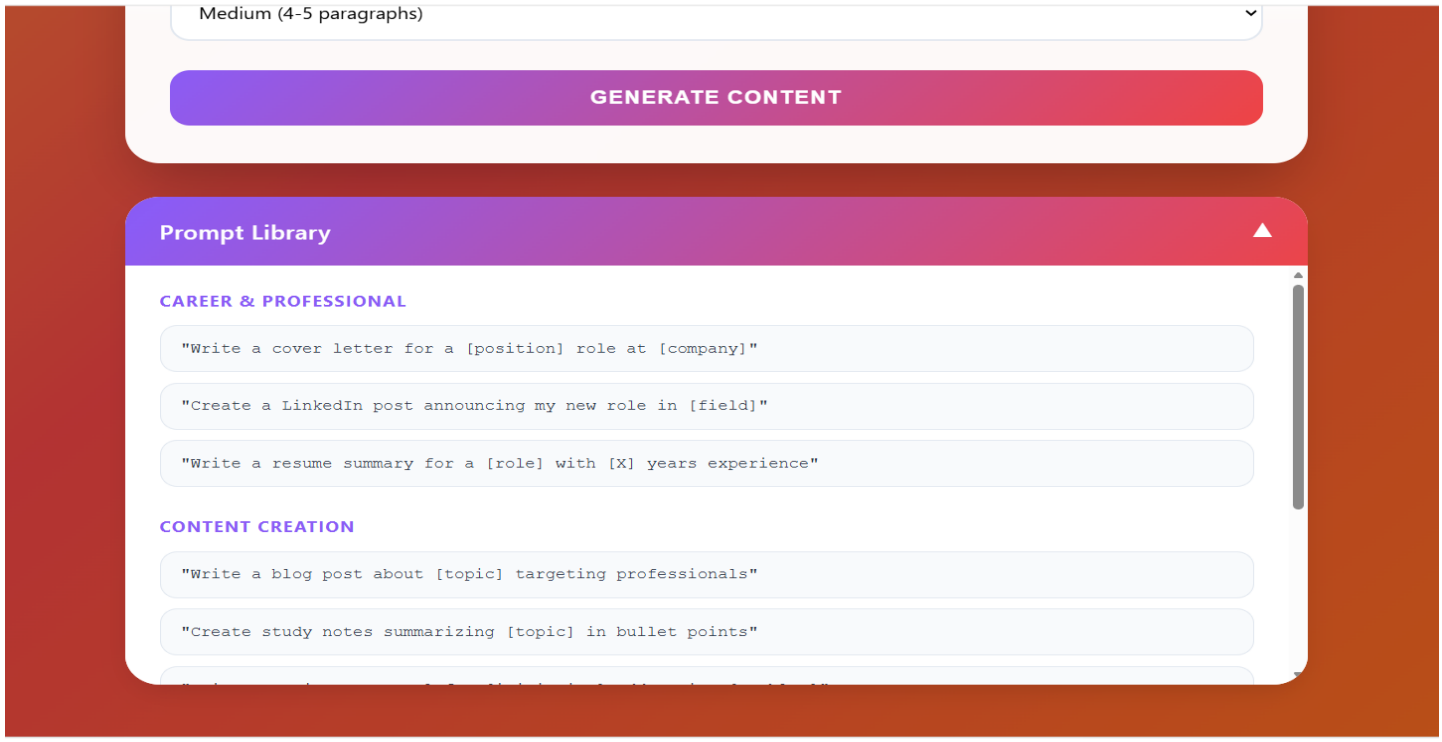


Figure 5 *— Prompt Library : Google Generative Language API (Gemini) endpoint used for inference.*

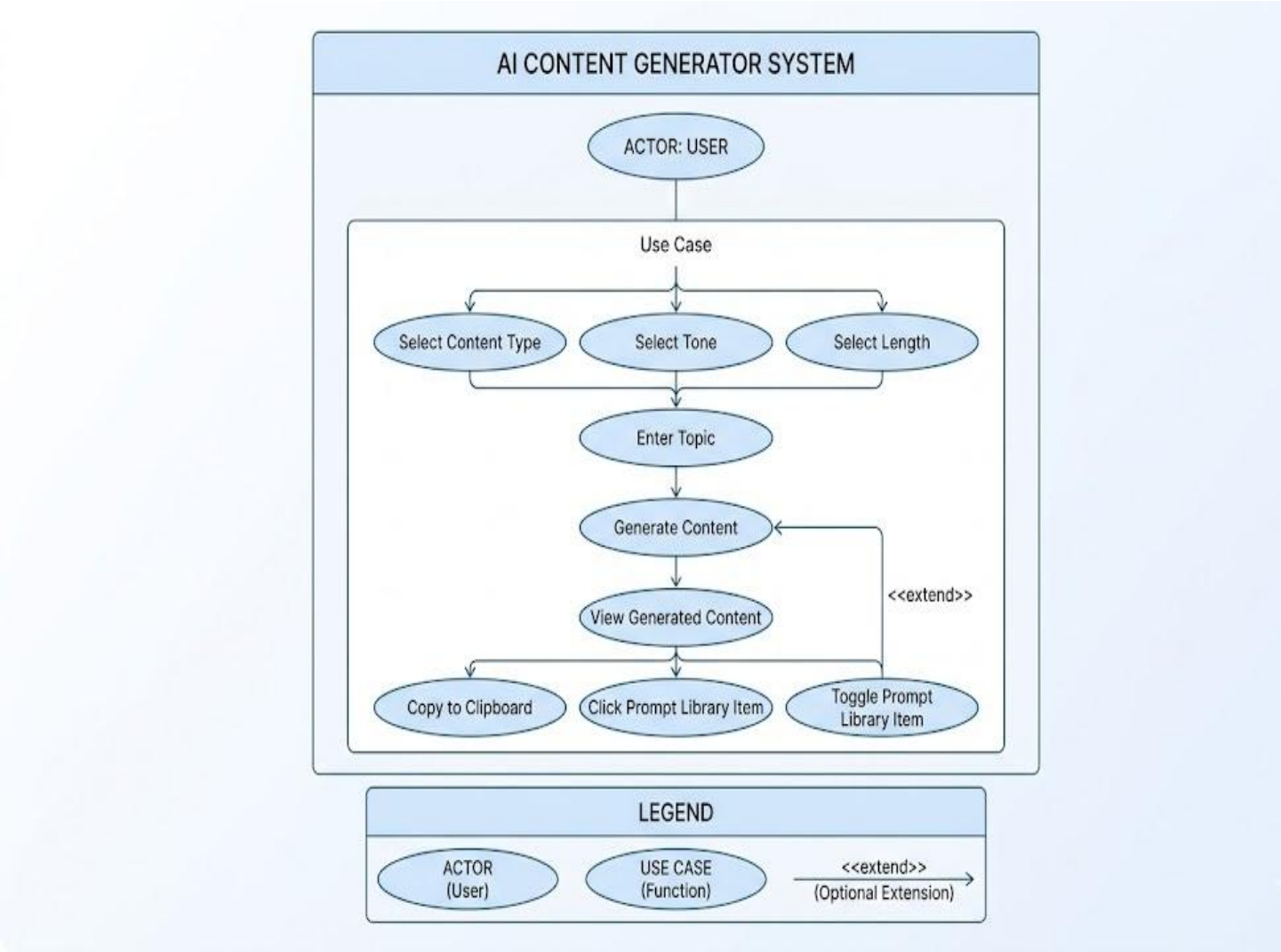
10. Reflection & Key Learnings

The most valuable lesson of the week was that prompt quality is the single biggest lever on output quality. Small changes, adding a role, specifying the audience, constraining the format, produced disproportionately better results than swapping models.

Building and deploying the project end to end also reinforced the importance of shipping. Putting the tool on a public URL forced decisions about UX, error handling, and how to communicate AI involvement honestly to users.

Going into Week 3, the priorities are: extend the prompt library with more domain-specific templates, add basic output validation, and start measuring quality against a small evaluation set rather than relying on subjective review.

11. UML Standard Use Case Diagram



12. Conclusion

Week 2 successfully delivered all required outcomes: a functional AI content generator, demonstrable prompt optimisation, completed coursework, and a published individual project. The work is live, documented, and integrated into the portfolio, and provides a strong foundation for the more advanced AI integration work scheduled in the weeks ahead.